

PROBABILITY THEORY

MM 4

MM 4: Special probability distributions

Topics: Discrete distributions:
Bernoulli, binomial, geometric, Poisson

Quizz

- Can the following be negative:
 - A r.v. X
 - $E[X]$
 - $E[X^2]$
 - $\text{Var}(X)$
 - $\text{Cov}(X, Y)$

What should we learn today?

- A number of r.v.s. arises in many diverse, unrelated applications → learn them
- What are the main facts about these distributions:
 - Their cdf and pdf
 - Mean and variance
 - Moment generating function
- In which situations the special distributions arise and how are they interrelated?

Bernoulli r.v.

- An experiment with 2 possible outcomes (success or failure) is called a Bernoulli trial.
- The indicator of event A is called the Bernoulli r.v., since it describes outcome of a Bernoulli trial.

$$E[I] = p$$

$$Var(I) = p(1 - p)$$

- Every Bernoulli trial, regardless of the definition of A, is equivalent to the tossing of a biased coin.

Example: indicator of an event

- **Example:** a r.v. I is an indicator of the event A
- *The expected value of the indicator of an event is equal to the probability of the event*

$$I = \begin{cases} 1, & \text{if } A \text{ occurs} \\ 0, & \text{if } A \text{ does not occur} \end{cases}$$

$$E[I] = 1 \cdot P(A) + 0 \cdot P(A^c) = P(A)$$

Example: indicator of an event

$$I = \begin{cases} 1, & \text{if event } A \text{ occurs} \\ 0, & \text{if event } A \text{ does not occur} \end{cases}$$

$$\text{Var}(I) = E[I^2] - E[I]^2 = E[I] - E[I]^2 =$$

$$I^2 = I$$

$$= E[I](1 - E[I]) = P(A)(1 - P(A)) = P(A)P(A^c)$$

Binomial r.v.

- Let X be a r.v. representing the number of times a certain event A occurs in n trials (number of success in n trials). Denote p probability of success.
- Then X is said to be a binomial r.v. with parameters n and p

$$X \sim b(n, p)$$

- Pmf

$$P\{X = i\} = \binom{n}{i} p^i (1 - p)^{n-i}, \quad n = 0, 1, \dots, n$$

$$E[X] = np$$

$$Var(X) = np(1 - p)$$

Binomial r.v.: mean and variance

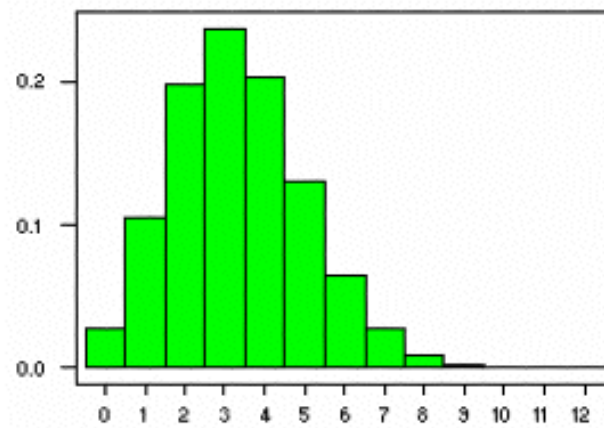
$$X = I_1 + I_2 + \dots + I_n$$

where I_i is a Bernoulli r.v.

$$\begin{aligned} E[X] &= E[I_1 + I_2 + \dots + I_n] = E[I_1] + E[I_2] + \dots + E[I_n] = \\ &= \underbrace{p + p + \dots + p}_{n \text{ times}} = np \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= \text{Var}(I_1 + \dots + I_n) = \text{Var}(I_1) + \dots + \text{Var}(I_n) = \\ &= \underbrace{p(1-p) + \dots + p(1-p)}_{n \text{ times}} = np(1-p) \end{aligned}$$

Binomial r.v.



Parameters $n=20$, $p=1/6$

Computing the binomial cumulative distribution function

$$P\{X \leq i\} = \sum_{k=0}^i P\{X=k\} = \sum_{k=0}^i \binom{n}{k} p^k (1-p)^{n-k}$$

- To optimize the computational process, we can utilize the following relationship between $P\{X=k+1\}$ and $P\{X=k\}$:

$$P\{X = k + 1\} = \frac{p}{1-p} \frac{n-k}{k+1} P\{X = k\}$$

- Now starting from $P(X=0)$ we can recursively find all other $P(X=k)$

Example: n-component system

- A system consists of n components. Each component functions independently with probability p.
- Question: for which values of p is a 5-component system more likely to operate than a 3-component system?

For 5-component system:

$$\begin{aligned} P_1 &= P\{X \geq 3\} = P\{X=3\} + P\{X=4\} + P\{X=5\} = \\ &= 1 - P\{X < 3\} = 1 - P\{X=0\} - P\{X=1\} - P\{X=2\} = \\ &= \binom{5}{3} p^3 (1-p)^2 + \binom{5}{4} p^4 (1-p) + \binom{5}{5} p^5 (1-p)^0 \end{aligned}$$

Example (cntd)

For 3-component system:

$$\begin{aligned} P_2 &= P\{Y \geq 2\} = P\{Y=2\} + P\{Y=3\} = \\ &= \binom{3}{2} p^2(1-p) + \binom{3}{3} p^3(1-p)^0 \end{aligned}$$

We would like to choose p such that $P_1 \geq P_2$

$$\Rightarrow 10p^3(1-p)^2 + 5p^4(1-p) + p^5 \geq 3p^2(1-p) + p^3$$

$$3(1-p)^2(2p-1) \geq 0$$

$$p \geq \frac{1}{2}$$

- **Statement.** Let $X \sim \text{Binomial}(n, p)$ and $Y \sim \text{Binomial}(m, p)$ be two independent random variables. A new random variable is $Z = X + Y$. Z is a binomial random variable with parameters **$m+n$** and **p** , i.e., $\text{Binomial}(m+n, p)$.

- *Proof.*

X is the number of "success" in **n** independent Bernulli trials.

Y is the number of "success" in **m** independent bernulli trials.

$Z = X + Y$ will be the total number of "success" in **$n+m$** independent trials

Note, it works only if second parameter, p , is the same for X and Y

Why is it called binomial?

N / k	0	1	2	3	4
1	p	q			
2	p^2	$2pq$	q^2		
3	p^3	$3p^2q$	$3pq^2$	q^3	
4	p^4	$4p^3q$	$6p^2q^2$	$4pq^3$	q^4

Coefficients are the same as in the binomial expansion $(p+q)^n$

The geometric r.v.

- Let a r.v. M be a number of independent Bernoulli trials until the first occurrence of a success. M is called a geometric r.v.
- Pmf

$$P\{M = k\} = (1 - p)^{k-1}p$$

$$E[M] = \frac{1}{p} \qquad Var(M) = \frac{1 - p}{p^2}$$

- Example: applications where we are interested in the time that elapses between the occurrence of events in a sequence of independent experiments

Geometric r.v.: mean and variance

$$P\{M = k\} = (1 - p)^{k-1} p$$

$$E[M] = \sum_{k=1}^{+\infty} k p (1-p)^{k-1}$$

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$$

$$\sum_{k=0}^{+\infty} k x^{k-1} = \frac{1}{(1-x)^2}$$

$$x = 1-p$$

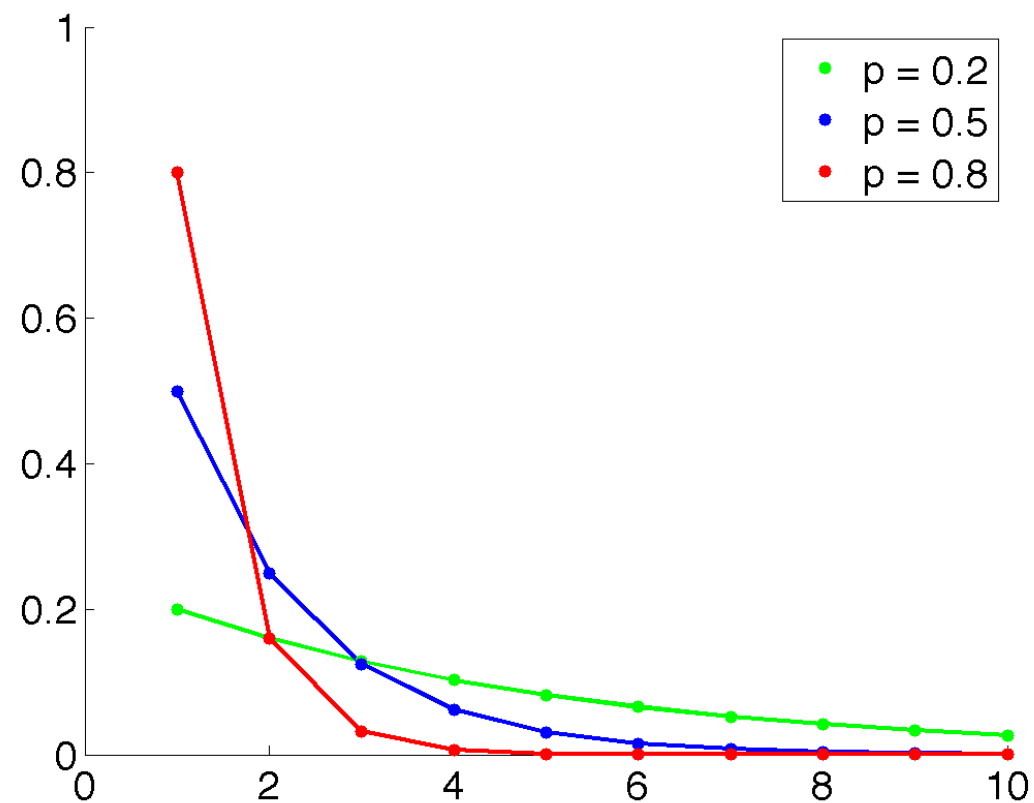
$$E[M] = p \cdot \frac{1}{(1-(1-p))^2} = \frac{p}{p^2} = \frac{1}{p}$$

$$E[M] = \frac{1}{p} \quad \text{Var}(M) = \frac{1-p}{p^2}$$

M	Probability
1	p
2	(1-p)p
3	(1-p) ² p
⋮	
k	(1-p) ^{k-1} p
⋮	

M ²	P _r
1	p
2 ²	(1-p)p
3 ²	(1-p) ² p
⋮	
k ²	(1-p) ^{k-1} p
⋮	

The geometric r.v.



Example: a communication system with infinite retransmissions

- A system retransmits a packet until it is received correctly
- $p=0.2$ is probability of failure
- How many times on average we will transmit a packet?
- X is geometrically distributed with parameter 0.8

Negative Binomial (Pascal) Distribution

- a generalization of the geometric distribution: independent trials are repeated until observing m successes.
- $X \sim \text{Pascal}(m, p)$
- $A = \{X = k\}$, and $A = B \cap C$, where
- B is the event that we observe $m-1$ successes in the first $k-1$ trials, C is the event that we observe a success in the k th trial.
- $P(A) = P(B \cap C) = P(B)P(C)$

$$P(B) = \binom{k-1}{m-1} p^{m-1} (1-p)^{\overbrace{(k-1)-(m-1)}^{k-m}}$$

$$P(X = k) = \binom{k-1}{m-1} p^m (1-p)^{k-m}$$

The hypergeometric r.v.

- Suppose we have objects of two types: type 1 N objects (acceptable components) and type 2 M objects (defective components). A sample of n objects is randomly chosen without replacement.
- A r.v. X represents a number of type 1 objects in the selection $\rightarrow$ hypergeometric with parameters (N, M, n)
- Pmf

$$P\{X = i\} = \frac{\binom{N}{i} \cdot \binom{M}{n-i}}{\binom{N+M}{n}}$$

$$E[X] = \frac{nN}{N+M} \quad \text{Var}(X) = np(1-p) \left[1 - \frac{n-1}{N+M-1} \right]$$

Derivation of formulas for mean and variance of hypergeometric r.v.

$$X = I_1 + I_2 + \dots + I_n \quad P\{I_k = 1\} = \frac{N}{N+M} = p$$

$$E[X] = \sum_{k=1}^n E[I_k] = np = \frac{nN}{N+M}$$

I_k are not independent $\Rightarrow$

$$\text{Var}(X) = \sum_{k=1}^n \text{Var}(I_k) + 2 \sum_{1 \leq k < l \leq n} \text{Cov}(I_k, I_l)$$

$$\text{Var}(I_k) = p(1-p)$$

$$\text{Cov}(I_k, I_l) = E[I_k I_l] - E[I_k]E[I_l]$$

$$\begin{aligned} E[I_k I_l] &= P\{I_k I_l = 1\} = P\{I_k = 1, I_l = 1\} = \\ &= \frac{N}{N+M} \cdot \frac{N-1}{N+M-1} \end{aligned}$$

Poisson r.v.

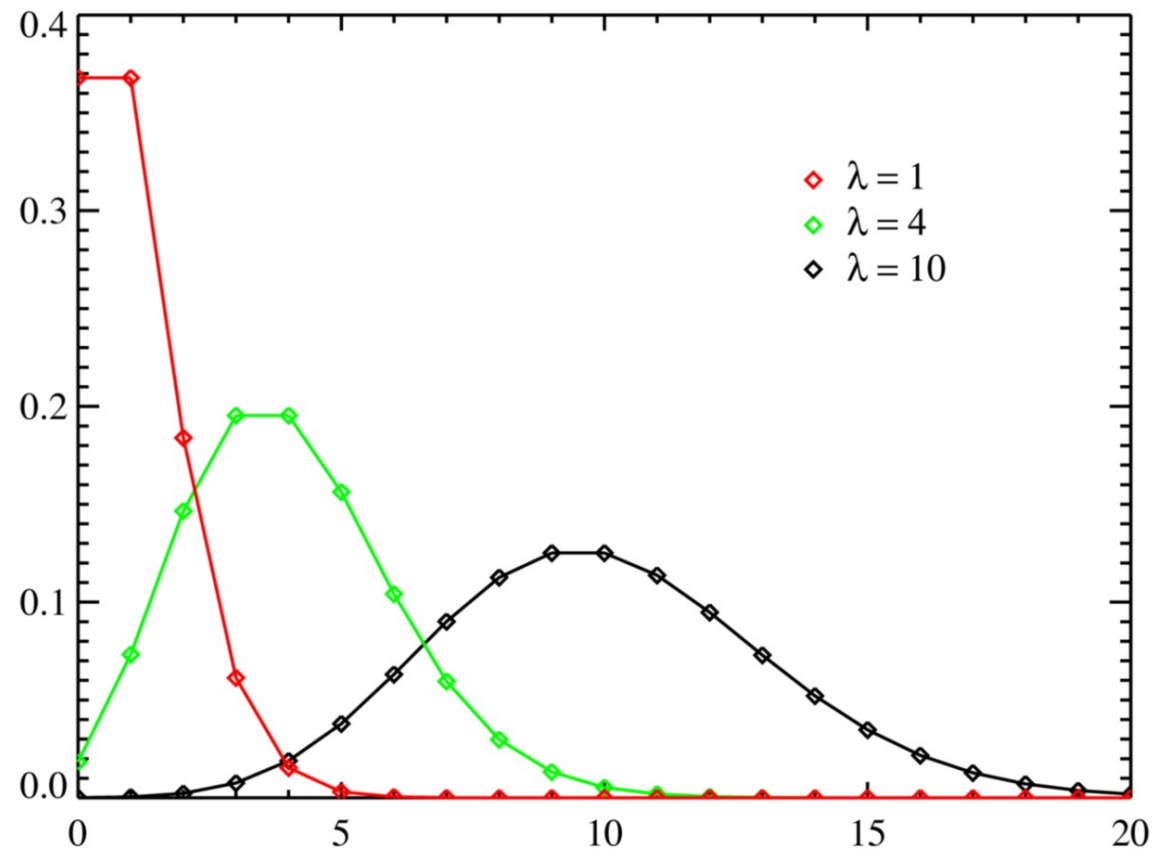
- A r.v. taking on one of values 0, 1, 2, ... is said to be a Poisson r.v. with parameter λ ($\lambda > 0$), if pmf is given by

$$P\{X = i\} = e^{-\lambda} \frac{\lambda^i}{i!}$$

$$E[X] = \lambda$$

$$Var(X) = \lambda$$

Poisson r.v.

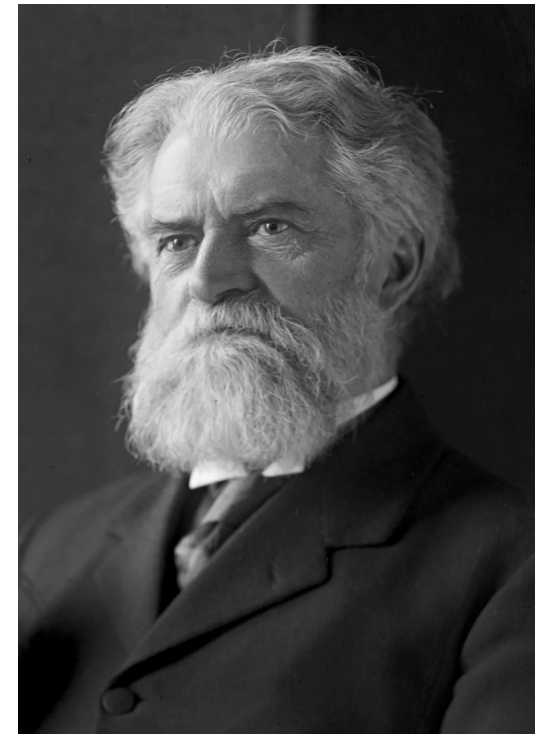


Examples

- Examples of events that can be modelled by Poisson distribution:
 - Number of typos (misprints) on a page of a book
 - Number of phone calls at a call center per minute
 - Number of times a web server is accessed per minute
 - Number of pine trees per unit area in a mixed forest
 - Number of stars in a given volume of space
 - ...



1837 Poisson; a number of wrongful convictions in a court



1860 Simon Newcomb; a number of stars found in an unit of space



1898 Ladislaus Bortkiewicz; the
number of soldiers in the Prussian
army killed accidentally by horse
kicks

When is the Poisson distribution an appropriate model?

- The occurrence of one event does not affect the probability that a second event will occur. That is, events occur independently.
- The rate at which events occur is constant. The rate cannot be higher in some intervals and lower in other intervals.
- The probability of an event in a small interval is proportional to the length of the interval.

Approximation for a binomial r.v.

- One of the applications of the Poisson probabilities is to approximate the binomial probabilities – when n is large and p is small.
- $\lambda = np$

Let $X \sim b(n, p)$. We put $\lambda = np$

$$P\{X=i\} = \frac{n!}{(n-i)!i!} p^i (1-p)^{n-i} = \frac{n!}{(n-i)!i!} \left(\frac{\lambda}{n}\right)^i \left(1 - \frac{\lambda}{n}\right)^{n-i} =$$

$$= \underbrace{\frac{n(n-1)\dots(n-i+1)}{n^i}}_{\xrightarrow{n \rightarrow \infty} 1} \cdot \frac{\lambda^i}{i!} \cdot \frac{\underbrace{\left(1 - \frac{\lambda}{n}\right)^n}_{\xrightarrow{n \rightarrow \infty} e^{-\lambda}}}{\underbrace{\left(1 - \frac{\lambda}{n}\right)^i}_{\xrightarrow{n \rightarrow \infty} 1}} \rightarrow e^{-\lambda} \cdot \frac{\lambda^i}{i!}$$

Properties of Poisson distribution

- The probability of at least one occurrence of the event in a given time interval is proportional to the length of the interval.
- The probability of two or more occurrences of the event in a very small time interval is negligible.
- The numbers of occurrences of the event in disjoint time intervals are mutually independent.

Mean and variance of Poisson distribution

$$\begin{aligned} E[X] &= \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda} \lambda^k}{k!} = e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} = \\ &= e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^{j+1}}{j!} = e^{-\lambda} \cdot \lambda \cdot \underbrace{\sum_{j=0}^{\infty} \frac{\lambda^j}{j!}}_{e^{\lambda}} = e^{-\lambda} \cdot \lambda \cdot e^{\lambda} = \lambda \end{aligned}$$

$$E[X] = \lambda$$

no derivations, just formula for variance:

$$\text{Var}(X) = \lambda$$

Example

- The BER of a communication channel is 10^{-3} . What is a probability that a block of 1000 bits has five or more errors?

Using Binomial distribution:

$$\begin{aligned} P\{X \geq 5\} &= 1 - P\{X < 5\} = \\ &= 1 - \sum_{k=0}^4 \binom{1000}{k} (10^{-3})^k (1-10^{-3})^{1000-k} = \\ &= 1 - 1 \cdot 1 \cdot (1-10^{-3})^{1000} - 1000 \cdot (10^{-3}) (1-10^{-3})^{999} - \\ &\quad - 500 \cdot 999 \cdot (10^{-3})^2 (1-10^{-3})^{998} - 500 \cdot 333 \cdot 998 \cdot (10^{-3})^3 (1-10^{-3})^{997} - \\ &\quad - 250 \cdot 333 \cdot 998 \cdot 997 \cdot (10^{-3})^4 (1-10^{-3})^{996} \end{aligned}$$

Example

- Using Poisson approximation we get:

$$X \sim b(n, p) \quad n = 1000, \quad p = 10^{-3}$$

Poisson approximation: $\lambda = np = 1$

$$\begin{aligned} P\{X \geq 5\} &= 1 - P\{X < 5\} = 1 - \sum_{k=0}^4 \frac{\lambda^k}{k!} e^{-\lambda} = \\ &= 1 - e^{-1} \left(1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} \right) \approx 0.0036 \end{aligned}$$

Properties of a Poisson r.v.

- Reproductive property
- Proposition.
 - 1) The sum of independent Poisson r.v.s. with parameters λ_1 and λ_2 is also a Poisson r.v. with parameter $\lambda_1 + \lambda_2$
 - 2) If a Poisson event having mean λ can be independently classified as being of type 1, 2, ... r with respective probabilities

$$p_1, p_2, \dots, p_r \left(\sum p_i = 1 \right)$$

then type i events are independent Poisson r.v. with means λp_1 , $\lambda p_2 \dots \lambda p_r$

Distributions:

- Bernoulli
- Binomial
- Geometric
- Pascal
- Hypergeometric
- Poisson

- Problem 1. Suppose that an average number of accidents on a busy crossing occurring during a month is 3. Calculate probability that there is at least one accident per month.
- Problem 2. Components produced at a factory are defective with probability 0.1. They are sold in package of 20 componets and a return is accepted, if more than one component is defective in a package. What is a probability that a package is returned?
- Problem 3. A deck of cards contains 20 cards: 6 red cards and 14 black cards. 5 cards are drawn randomly *without replacement*. What is the probability that exactly 4 red cards are drawn?
- Problem 4. It is known that 20% of products on a production line are defective. Products are inspected until first defective is encountered. How many inspections will be done to obtain the first defective product?